

SAI SUDHA G

DATA ENGINEER

Sudhag9422@gmail.com | +1(972)7680034 | [Linkedin](#) | [Github](#)

PROFESSIONAL SUMMARY

- Data Engineer with 5+ years of experience designing, building, and maintaining large-scale data platforms across cloud-native environments including Azure, AWS, and GCP.
- Deep expertise in Apache Spark and Databricks for distributed data processing, real-time streaming, and complex batch pipeline orchestration at enterprise scale.
- Proficient in building end-to-end ELT/ETL workflows using Azure Data Factory, dbt, Airflow, and Informatica, covering ingestion, transformation, and delivery layers.
- Strong foundation in data lakehouse architecture using Delta Lake, Apache Hudi, and Iceberg formats with ACID compliance, time travel, and schema enforcement.
- Experienced in cloud data warehousing with Snowflake, Google BigQuery, Azure Synapse, and Amazon Redshift — including query optimization, clustering, and cost management.
- Advanced SQL practitioner with deep knowledge of window functions, CTEs, query tuning, execution plans, and performance optimization across distributed systems.
- Skilled in designing relational data models, implementing SQL Server indexing strategies, and writing stored procedures for production-grade data transformations.
- Proven track record leading cross-functional data engineering teams with full ownership of complex migrations, infrastructure deployments, and enterprise data governance initiatives.

TECHNICAL SKILLS

Big Data & Distributed Processing: Apache Spark, Databricks, Kafka, Hadoop, Hive, Apache Flink, Parquet, ORC, Avro, Delta Lake, Apache Hudi, Iceberg

Cloud – AWS: Glue, EMR, Lambda, S3, Redshift, Kinesis, Athena, Step Functions, CloudWatch

Cloud – Azure: Azure Data Factory, Databricks, Synapse Analytics, Azure Monitor, ADLS Gen2, Event Hubs, Azure DevOps

Cloud – GCP: DataProc, BigQuery, GCS, Pub/Sub, Cloud Composer, Dataflow

Relational Databases & Data Warehousing: SQL Server, Snowflake, Google BigQuery, Oracle, PostgreSQL, MySQL, DynamoDB, Cosmos DB

SQL & Query Excellence: Advanced Joins, CTEs, Window Functions, Query Optimization, Execution Plans, SQL Server Indexing Strategies, Stored Procedures

Programming & Scripting: Python, Pandas, NumPy, Scikit-learn, PySpark, Scala, MATLAB, Bash

ETL & Pipeline Orchestration: dbt, Apache Airflow, Azure Data Factory, Informatica, SSIS, SSAS, Fivetran, Talend

Data Modeling & Architecture: Lakehouse, Lambda, Kappa, Medallion (Bronze/Silver/Gold), Data Mesh, Event-Driven, Dimensional Design, Star/Snowflake Schemas

BI & Visualization: Power BI, Tableau, QuickSight, Looker, Grafana

Data Quality & Governance: Great Expectations, Monte Carlo, Custom Validation Frameworks, Schema Contracts, Anomaly Detection, Data Lineage, Metadata Management

DevOps & Infrastructure: Git, GitHub Actions, Jenkins, Docker, Kubernetes, Terraform, Ansible, Jira, ServiceNow, Linux, CI/CD Pipelines

ML & AI Support: Feature Stores, Training Dataset Pipelines, MLflow Integration, Model Serving Data Layers

WORK EXPERIENCE

Meta, Data Engineer

Mar 2025 – Present

Dallas, Texas

- Architected and delivered enterprise-scale data pipelines using Oracle PL/SQL, ODI, Databricks, and dbt, supporting analytics, reporting, and AI workloads across multiple domains.

- Designed and implemented a modular lakehouse architecture on Databricks with Delta Lake, enabling ACID-compliant transactions, time travel, and schema enforcement across critical data domains.
- Built robust incremental and CDC-based loading strategies to replace full-load pipelines, significantly reducing processing time and compute costs across high-volume ingestion workflows.
- Developed reusable dbt model libraries with standardized transformation patterns, macros, and documentation, accelerating feature delivery and enforcing consistency across data engineering teams.
- Integrated data from heterogeneous source systems including relational databases, SaaS APIs, event streams, and flat-file feeds into a unified data lakehouse using orchestrated batch and streaming pipelines.
- Engineered a comprehensive data quality framework with schema validation, referential integrity checks, null handling, and anomaly detection to ensure production-grade data trustworthiness.
- Implemented data lineage tracking and metadata cataloging using enterprise data governance tools, enabling full audit trails from source ingestion through BI and ML consumption.
- Designed semantic data models and curated datasets powering executive dashboards in Power BI, enabling self-service analytics and reducing ad-hoc engineering requests.
- Collaborated with applied scientists and ML engineers to build feature pipelines and training datasets for personalization and recommendation platforms.
- Established performance monitoring and alerting using structured logging, pipeline health dashboards, and SLA tracking to ensure reliability and proactive incident resolution.
- Led query optimization efforts including partition pruning, Z-ordering, broadcast joins, and caching strategies to improve pipeline throughput on large-scale Spark workloads.
- Implemented CI/CD pipelines for data engineering using Git-based workflows, automated testing, and deployment automation, enabling frequent and safe production releases.
- Defined and enforced data contracts between producing and consuming teams, reducing schema-breaking incidents and improving cross-team data reliability.
- Partnered with data governance and security teams to classify sensitive data, enforce role-based access controls, and ensure compliance with enterprise data privacy policies.
- Drove adoption of infrastructure-as-code practices using Terraform to provision and manage cloud resources, eliminating manual configuration and ensuring environment consistency.
- Translated ambiguous business problems into concrete technical solutions, designing data models and pipelines that addressed stakeholder needs with clear documentation and proof-of-concept validation.
- Owned full responsibility for data pipeline reliability and performance, proactively identifying bottlenecks and implementing solutions from source ingestion through downstream consumption layers.

Horizon BCBSNJ, Data Engineer II

Jan 2023 – Dec 2023

Hyderabad, India

- Led end-to-end migration of a legacy on-premises BI platform to Azure, delivering Databricks notebooks in PySpark and SQL with full data fidelity validation and zero production discrepancies.
- Redesigned batch pipeline architecture through Spark optimization techniques including partition tuning, skew handling, predicate pushdown, and elimination of redundant processing stages.
- Implemented Delta Lake as the core storage layer, introducing ACID transactions, merge-based upserts, and time travel capabilities that replaced fragile overwrite-based patterns.
- Established unit testing, integration testing, and peer review workflows for all pipeline code, dramatically improving reliability and reducing production incidents across the platform.
- Designed and deployed a performance monitoring stack using Azure Monitor, custom logging, and alerting rules, enabling proactive detection and resolution of pipeline bottlenecks.
- Achieved significant cloud cost reductions by right-sizing Databricks clusters, transitioning to spot and autoscaling instances, and optimizing Delta Lake file compaction and retention policies.
- Built reusable PySpark transformation libraries and shared utility modules, standardizing data processing logic and enabling rapid development of new pipeline features.
- Developed a metadata-driven pipeline framework that allowed business teams to onboard new data sources without requiring custom engineering effort for each integration.
- Created comprehensive data dictionaries, pipeline architecture diagrams, and operational runbooks, establishing institutional knowledge and reducing onboarding time for new engineers.
- Implemented schema evolution handling and backward-compatible migration strategies, ensuring pipeline resilience when upstream source schemas changed without engineering intervention.
- Delivered curated analytical datasets and aggregate tables into Azure Synapse, enabling fast query performance for BI tools and reducing load on operational source systems.

- Led a cross-functional engineering team in Agile sprints, owning backlog management, CI/CD pipeline delivery, and stakeholder communication across technical and business groups.
- Drove migration strategy by establishing data validation frameworks and reconciliation dashboards that verified functional correctness and business rule compliance post-migration.
- Mentored junior engineers on SQL Server indexing strategies, stored procedure design, and relational database best practices to build team depth in query optimization.

Cognizant Technology Solutions, Data Engineer

Jan 2020 – Dec 2022

Hyderabad, India

- Designed and operated production ELT pipelines using Azure Data Factory and Databricks, processing millions of records across multiple dependent systems on high-frequency recurring schedules.
- Developed a Python and Spark SQL data validation framework that systematically eliminated data errors, reduced manual verification effort, and built confidence in downstream consumption.
- Built and maintained complex ADF pipelines with parameterized templates, dependency chaining, error handling, and retry logic, ensuring robust and observable data delivery.
- Maintained multiple production-grade data platforms with high availability SLAs, performing root cause analysis and implementing permanent fixes for recurring pipeline failures.
- Created an extensive SQL view and stored procedure library enabling self-service data validation and exploration for business stakeholders without requiring engineering intervention.
- Collaborated with data analysts and BI developers to model, cleanse, and deliver trusted datasets powering reporting dashboards and executive decision-making workflows.
- Developed comprehensive SQL Server indexing strategies and optimized stored procedures to improve query performance, reducing dashboard load times for end-user reporting.
- Implemented data archival and retention policies aligned with business requirements, optimizing storage costs while maintaining accessibility for historical reporting needs.

EDUCATION

University of Central Missouri, MO, USA

Master of Science in Computer Science

JNT University Kakinada, India

Bachelor of Science in Computer Science